

Note first that the set $A = [\ell_1, u_1] \times [\ell_2, u_2]$ is non empty and convex. Consider x and y in A and $\lambda \in [0, 1]$. By convexity of A , the point $z = (1 - \lambda)x + \lambda y$ belongs to A .

As f is concave, we have

$$\begin{aligned} g(z) &= f(z_1) \\ &= f((1 - \lambda)x_1 + \lambda y_1) \\ &\geq (1 - \lambda)f(x_1) + \lambda f(y_1) \\ &= (1 - \lambda)g(x_1, x_2) + \lambda g(y_1, y_2) \\ &= (1 - \lambda)g(x) + \lambda g(y), \end{aligned}$$

that proves the concavity of g .